

Mohammad Khan

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience building and deploying production-grade full-stack and AI/ML systems. Strong in **Python, Java, React, AWS, AI agents, RAG, and modern LLM tooling**, with experience designing scalable backend services and cloud architectures. Seeking a **full-stack, backend, or AI/ML engineering** role where I can contribute immediately, solve real-world problems, and continue growing as an engineer.

TECHNICAL SKILLS

AI/ML & GenAI: RAG, LangGraph, LLM Agents, Hugging Face, OpenAI API, Claude Code, GitHub Copilot, Cursor
Languages: Java, Python, JavaScript, TypeScript, C, C++
Frontend: React, Next.js, Angular, TailwindCSS, HTML/CSS
Backend: Node.js, FastAPI, Express.js, Spring Boot, .NET, Django, REST APIs
Cloud & DevOps: AWS, Docker, Git, Kubernetes, Terraform, Linux, GitHub Actions (CI/CD)
Databases: PostgreSQL, MongoDB, MySQL, NoSQL, pgvector, DynamoDB
Networking & Systems: TCP/IP, DNS, HTTP/HTTPS, Load Balancing, VPNs, Linux Administration, Network Troubleshooting, Client-Server Architecture
Certifications: AWS Certified Cloud Practitioner, AWS Certified Developer - Associate, Kubernetes and Cloud Native Associate (KCNA), HashiCorp Terraform Associate, LPI Linux Essentials, Fortinet NSE 1-3 Certified in Cybersecurity

EXPERIENCE

Software Developer (Co-Op)

Jan. 2022 – Aug. 2022

Tread

Toronto, ON

- Reduced manual scheduling effort by **50%** by developing a **React-based** calendar management system with **REST API** integrations, automating subcontractor dispatch and availability tracking
- Eliminated payroll disputes by building reusable timesheet dashboard components in **React**, integrated with backend payroll services for accurate time tracking
- Improved post-release stability by identifying and resolving bugs through structured **QA** testing and **Jira-tracked** defect workflows

TECHNICAL PROJECTS

VeriNews AI - AI Fake News Detector | [Demo](#)

Jun. 2025 – Apr. 2026

AI/ML Engineering | Python, FastAPI, React, Node.js, MongoDB

- Achieved **92%** classification accuracy on **50,000+** news articles by fine-tuning Meta AI's **RoBERTa-base** model with **PyTorch** and **Hugging Face Transformers** to enable real-time misinformation detection
- Designed and deployed a three-service cloud architecture (**React, Node.js, FastAPI**) across **Vercel** and **Render**, enabling reliable and scalable AI inference for real-time article analysis
- Improved model transparency by integrating **LIME** explainability outputs and a **MongoDB** feedback loop for continuous model retraining

VeloRoute - Full-Stack GPS Analytics Engine | [Demo](#)

Jun. 2026 – Aug. 2026

Full-Stack & Cloud Engineering | Java, Spring Boot, PostgreSQL, AWS

- Developed an interactive geospatial dashboard that animated **10,000+** GPS coordinates with synchronized real-time ride metrics, by building a **React** and **Leaflet.js** frontend for live route playback
- Reduced API response time to under **50 ms** for large GPS file uploads, by designing an asynchronous **Spring Boot** pipeline with **PostgreSQL** persistence and **JSON** streaming to enable efficient metric processing
- Enabled cloud deployment by provisioning **AWS EC2** for **Spring Boot** backend, **Amazon RDS PostgreSQL** for data persistence, and **AWS S3** for **React** frontend hosting to deliver cloud-hosted GPS analytics platform

EDUCATION

Carleton University

Ottawa, ON

Honours Bachelor of Computer Science, Stream: Computer & Internet Security

Sept. 2019 – Jun. 2024

- Awards: Dean's Honour List (2020-2022), University Entrance Scholarship, George Fierheller Scholarship, Gerhard Herzberg Scholarship
- Related Coursework: Object-Oriented Programming, Computing in Python, Data Structures and Algorithms, Database Management Systems, Software Engineering Practices